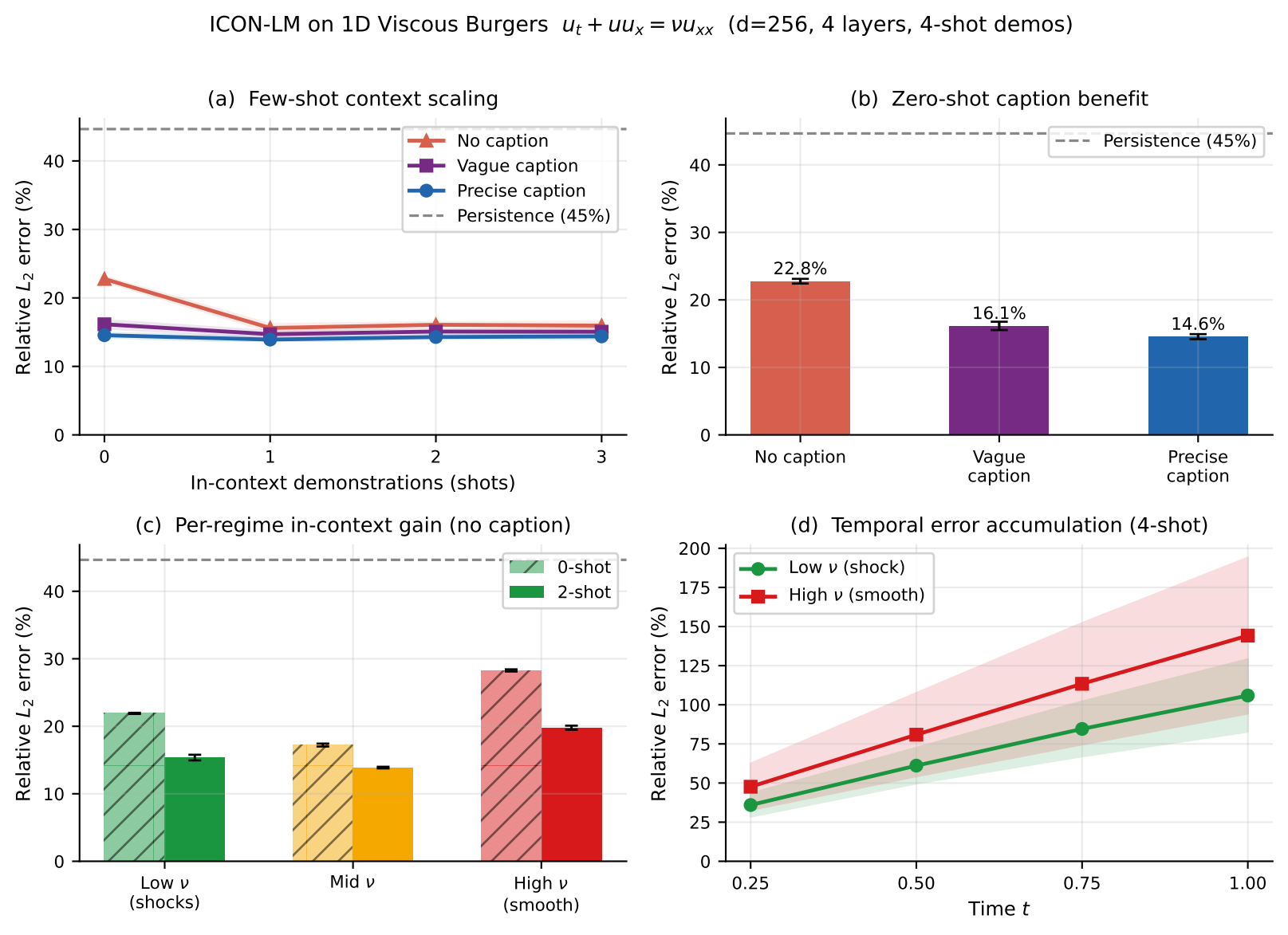
ICON-LM on 1D Viscous Burgers $u_t + uu_x = \nu u_{xx}$ (d=256, 4 layers, 4-shot demos)
(a) Few-shot context scaling
Relative L_2 error (%)
In-context demonstrations (shots)
No caption
Vague caption
Precise caption
Persistence (45%)
(b) Zero-shot caption benefit
Relative L_2 error (%)
Persistence (45%)
22.8%
16.1%
14.6%
No caption
Vague caption
Precise caption
(c) Per-regime in-context gain (no caption)
Relative L_2 error (%)
0-shot
2-shot
Low ν (shocks)
Mid ν
High ν (smooth)
(d) Temporal error accumulation (4-shot)
Relative L_2 error (%)
Low ν (shock)
High ν (smooth)
Time t